

TRI

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□ □ □ □ □ Updating the World Is Not Rebinding the Target: Temporal Referent Integrity in Tool-Using Agents

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“ ” “ ”

Temporal Referent Integrity TRI

SQLite

TRI

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API

ID

A

B

Agent

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A

ID

“

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TRI

- 1. “ ” “ ”
- 2.
- 3.



2.1



2.2



2.3



3.1

- 4.
- 5. TRI
- 6. gate
- 7. schema
- 8.
- 9. Agent benchmark

TRI

3.2

				S0			S1	
	schema							
	Flip			Stable			Remove	Invalidate
ID		Name Collision						
	160	20	8		240	40	10	schema
DeepSeek								Qwen GLM

3.3

				actor	gate		ID	
		reference mode			gate		Agent	
							CTA	
	Binding Drift				AWS Bedrock		re-verification	
	author adaptation			Binding Drift				

3.4

				TRI	Stable			TRI
	10,000	bootstrap		McNemar		API		intention-to-treat
		thinking		token				

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4.1

160			Qwen	GLM	64.4%	71.9%		95.0%	96.2%
gate	98.1%	100.0%			ID				

□ □ □		Qwen3.5	GLM-5.1
		64.4%	71.9%
		95.0%	96.2%
actor		96.9%	98.1%
+ gate		98.1%	100.0%

60.0% “ ” “ ”

4.2

/		75.0%	81.2%	70.6%	gate	65.0%	73.1%
CTA	actor	96.9%	98.1%	gate	1.2	1.9	
□□□□			Qwen3.5			GLM-5.1	
+ reference mode			75.0%			75.0%	
			81.2%			70.6%	
+ validity gate			65.0%			73.1%	
			95.0%			96.2%	
actor			96.9%			98.1%	
+ gate			98.1%			100.0%	
gate							

4.3 240

59/79		schema	Qwen GLM DeepSeek		43/72	38/80
		CTA	0			
□□		□□□□□□□□		CTA □□□□	CTA □□□□□□□□□□□□ □	
Qwen3.5		43/72		0/71	+23.3	
GLM-5.1		38/80		0/70	+24.2	
DeepSeek		59/79		0/70	+17.5	
		CTA				10
10,000		10		100%	20	100%
0						

4.4

ledger		240	Agent		TRI	
□□		□□□□□□	□□□□□□□□		CTA	
Qwen3.5		63.3%	69.6%		70.8%	
GLM-5.1		67.1%	80.8%		94.2%	
DeepSeek		68.8%	75.8%		91.2%	

Model	Qwen	DeepSeek	CTA	GLM
Qwen	1.2	0	56	38
DeepSeek	80	0	42	0
CTA	0	0	0	0
GLM	0	0	0	0

ID		40	SQLite	Qwen	GLM	13	8
gate	0	140					
	1,440	episode	Qwen	GLM	DeepSeek	87	79 75
70 46 57	“	”				validity gate	

gold Fleiss kappa 0.708 benchmark gold 86%

50 Qwen/GLM 60%/74% CTA 90%/98%

Preserve/Reevaluate fallback

80	schema				Qwen	46.2%	82.5%
	schema	selector grounding	40			guard	guard
					Qwen	35/40	39/40 GLM

Binding Drift		Entity Lock		TRI	
20	Entity Lock	10/10	0/10	GLM self-reverify	0/10 10/10
12/20	17/20	Qwen reverify	3/20	grounding	CTA Qwen GLM
prompt	TRI	verifier		“ ”	
Binding Drift	CTA	Binding Drift			

ToolSandbox	AppWorld	tau3-bench	ToolSandbox	129	
AppWorld	732	244	near-match tau3-bench	2,449	10,832
AppWorld	28	TRI			

Stable	TRI 0/2	ToolSandbox	benchmark	“ GLM	3/6
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Agent	TRI ToolSandbox	TRI “ ”	“ ” benchmark	LLM
Agent	ToolSandbox	“ ”	“ ” benchmark	LLM

“ ” “ ” TRI Binding Drift
Entity Binding Failures Agent

Binding Drift

160 240

“ benchmark ” TRI strict/near-match/excluded

CTA selector capability

TRI



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Binding Drift

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